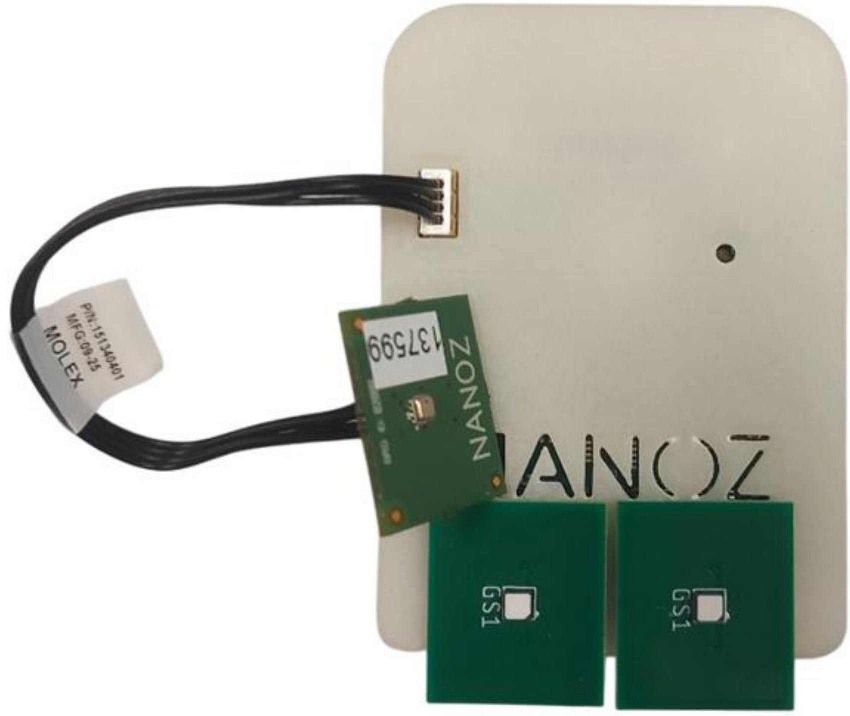
User manual for the software «Nanoz_EK»

Usable with Nanoz Evaluation Kit Generation IV



User Manual Revision 4.0.1, Software Nanoz_EK, Evaluation kit Generation IV

Contents

List of figures

List of tables

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I. Introduction

This user manual describes in detail the installation and the operation of the Evaluation Kit Generation IV (E.K.IV). It is composed of an electronic card in which two Nanoz chips can be plugged. These chips contain MOx sensors able to detect several gases (Ethanol, Formaldehyde, Acetone, BTEX, Carbon monoxide, Ozone…). The E.K.IV dimensions are 4.5x1.5x7cm^3.
It requires a micro-USB type C to fully function. It is recommended to use USB 3.0 or over. In the following pages, in depth instructions on the evaluation kit use will be explained. Complementary information will be mentioned as a *note.*

II. Sensors installation

Plug Nanoz gas and environmental sensors on the E.K.IV as shown in Figure 1. To ensure proper E.K.IV operation, chip positions as well as their plug-in directions should be respected.

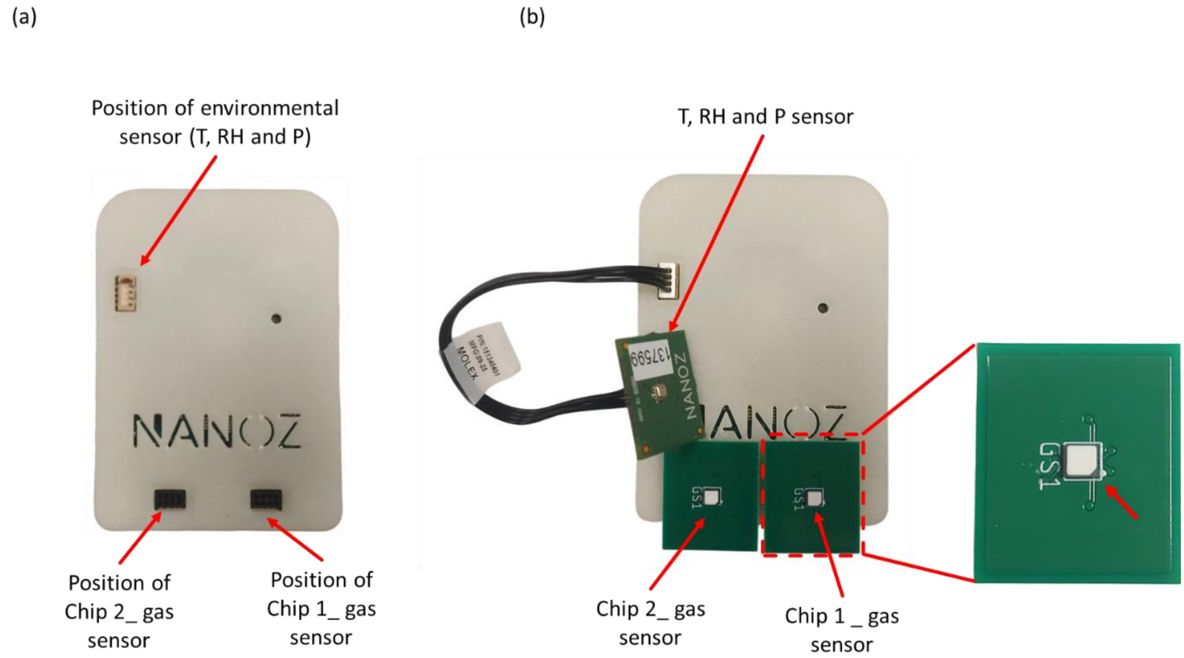

Figure 1: Nanoz Evaluation Kit Generation IV: (a) without gas and environmental sensors (b) with gas and environmental sensors.

Warning: To ensure proper operation and to prevent damage to the sensors, the maximum voltages of the sensors and heating elements to be applied must not exceed:

	Heater voltages (VH)	Sensor voltages (Vs)
NZGS2	2.2 V	+/- 0.8V
NZGS3	1.4V	+/- 0.5V

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III. Evaluation Kit software

Double click on the evaluation Kit software «Nanoz_EK». It has two main screens:

➔Configuration window for set up the evaluation kit (see Figure 2).
➔Control window for signal visualizations (see Figure 11).

The configuration window can be divided in six setting panels as described in Table 1.

	Name
A	Serial link
B	Configuration
C	Cycle
D	Sequence
E	E.K. communication management
F	Configuration / Control panel selection

Table 1 : Setting panels of the "Configuration" window.

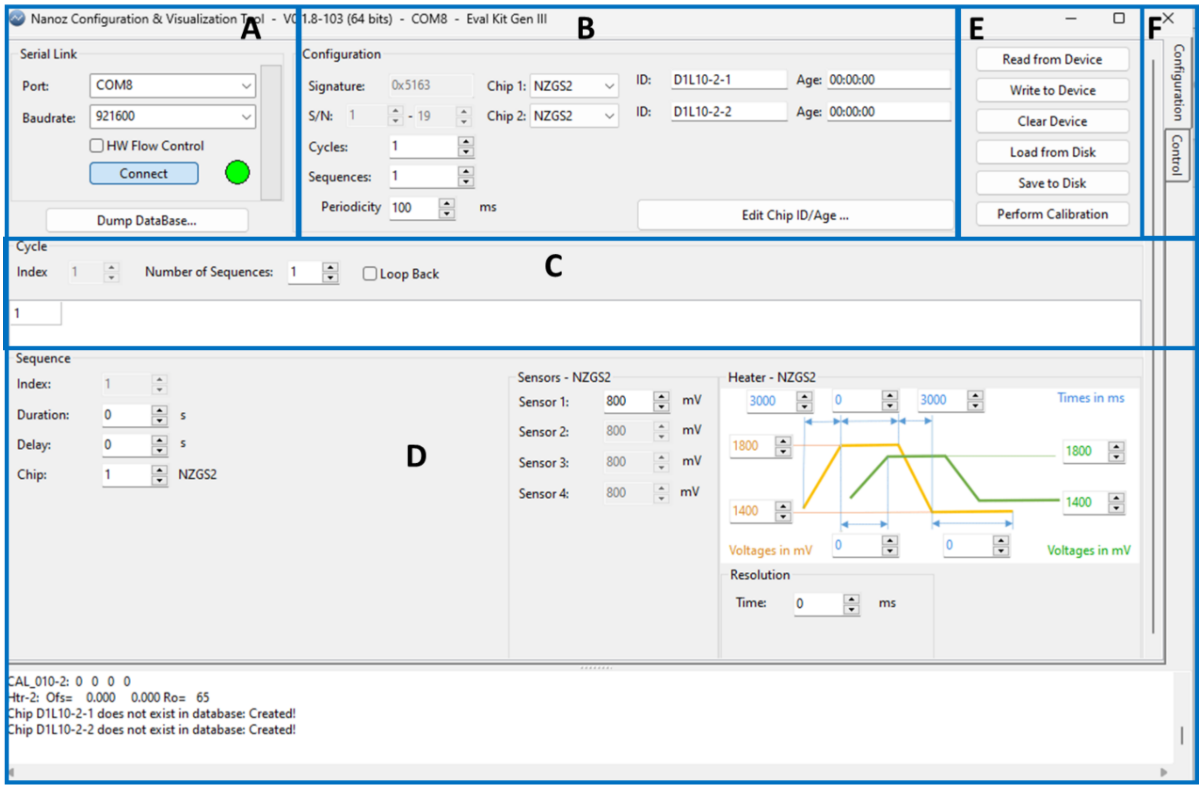

Figure 2 : Setting panels in the" Configuration" window.

IV. Configuration window

A. Serial link

Plug the evaluation kit on the USB-C port, select the corresponding « COM » and press «Connect». When it is done, the color marker on the software blinks green and yellow at first, then stays green.

This marker can have 3 states represented by 3 colors:

- Green: The evaluation kit is connected to the software.
- Yellow: The software is applying the instructions entered by the user.
- Red: The evaluation kit is disconnected from the computer.

The window presented on Figure 3 will be used to provide the names and ages of the sensors. Chip name could be found on the back side of the PCB to which the sensor is soldered. Chip age can be entered if known by the user. Otherwise, leave zero. This function has no impact on sensor operation and will be used when a gas prediction algorithm has been implemented. Press «Ok» to start configuration.

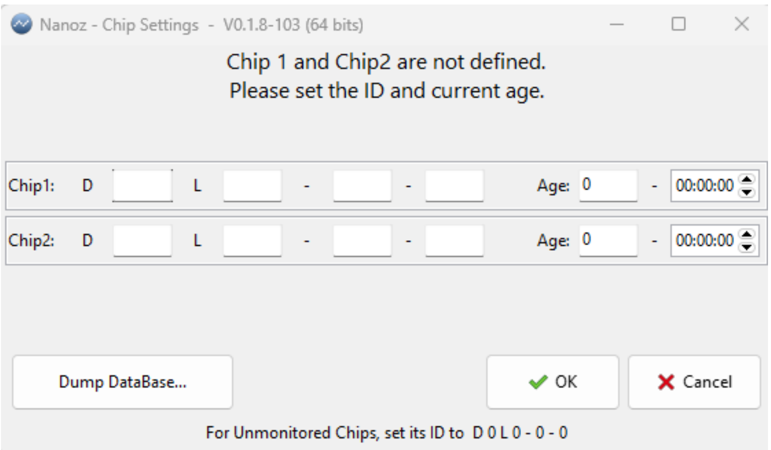

Figure 3: Window for name and age of chips.

B. Configuration

Figure 4: Configuration panel.

Cycles: Number of cycles that can be set up (from 1 to 48). One cycle can be composed of several sequences.

*Note: Each cycle is identified by its **index** value from 1 to 48.*

Sequences: Number of sequences that can be set up (from 1 to 96).

Note: One sequence contains a list of parameters which will be applied by the Evaluation kit. It includes for example the sensor and heater voltages, sequence duration….

Periodicity: Time between two measurement points in the same sequence (from 100 ms to 60000 ms).

Note: Other parameters must not be changed.

***Signature** and **S/N** are not modifiable and are used by Nanoz to identify the evaluation kit number.*

*The **S/N** corresponds to the E.K card number and will be appear in the recording file (Excel file).*

***Chip 1** and **Chip 2** are used to select the plugged chip.*

C. Cycle

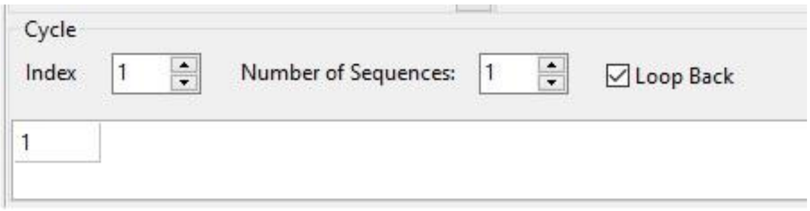

Figure 5: Cycle panel.

Index: cycle number.

Note: The cycle numbering starts by the index 1.

Number of sequences: Number of sequences in the active cycle. One cycle can be composed of several sequences.

Note: the number of sequences cannot exceed 29.

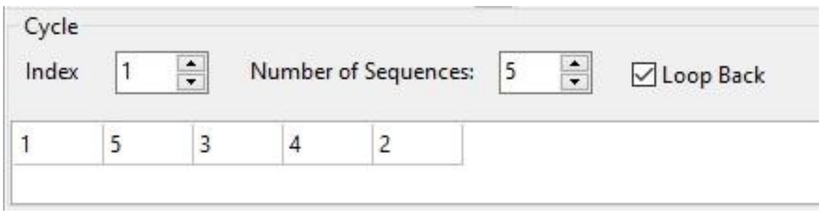

Figure 6: Example of one cycle with five sequences.

Warning: To ensure proper operation and to prevent damage to the sensors, the maximum voltages of the sensors and heating elements to be applied must not exceed:

	Heater voltages (VH)	Sensor voltages (Vs)
NZGS2	2.2 V	+/-0.8V
NZGS3	1.4V	+/-0.5V

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*There will be one box for each value entered in **Number of sequences**. The value in the box corresponds to the index sequence. It can be changed by double clicking in the box.*

Sequences can be executed in any order.

Loop back: When checked, the whole cycle starts again when it ends (infinitely).

D. Sequence

a) Sequence settings

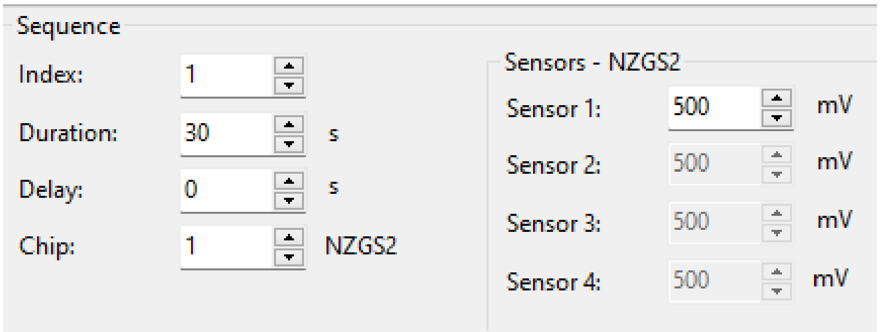

Figure 7: Sequence panel.

Index: Value used to identify a sequence.

Note: when there is only one sequence; the index is 1 and cannot be changed (the sequence number is the same defined in the Configuration panel).

Duration: Duration of the sequence in second (from 1 s to 60000 s).

Note: if the value is 0 second, the sequence will run for an infinite time.

Delay: Delay between two sequences in second (from 0 s to 60000 s).

Chip: Select the chip number.

Sensors-NZG2: Sensor voltage choice. One voltage can be applied for all sensors.

Note: Both chips (1 and 2) will run at the same time with the parameters defined previously.

Warning: To ensure proper operation and to prevent damage to the sensors, the maximum voltages of the sensors and heating elements to be applied must not exceed:

	Heater voltages (VH)	Sensor voltages (Vs)
NZGS2	2.2 V	+/-0.8V
NZGS3	1.4V	+/-0.5V

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b) Heater settings

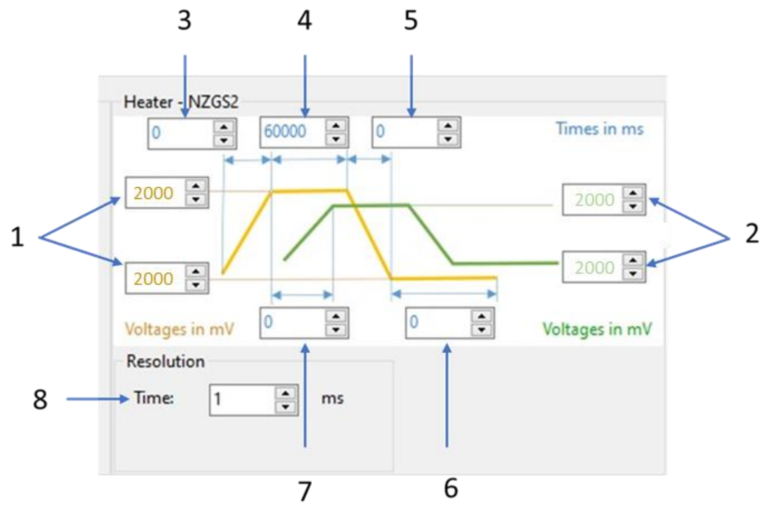

Figure 8: Heater settings panel.

	Parameters	Minimum	Maximum	Unit
1	Voltage for low & high states **Heater 1**	0	2200	mV
2	Voltage for low & high states **Heater 2**	0	2200	mV
3	Ramp up time	0	60000	ms
4	High state duration	0	60000	ms
5	Ramp down time	0	60000	ms
6	Low state duration	0	60000	ms
7	Phase shift between Heater 1 & 2	0	60000	ms
8	Time resolution	0	10000	ms

Table 2 : Heater control parameters.

Note: Each sequence and each chip have its own heating parameters.

The time resolution is used for to create a step function.

E. Evaluation Kit communication management

Figure 9: EK communication management panel.

Read from device: Read the last parameters saved in the E.K.IV card memory.

Write to device: Write all set up parameters from the Heater settings panel to the E.K.IV.

Note: Please note that all the modified values in the configuration window will be applied only after the « Write to Device » has been done.

Clear device: Erase the settings currently saved on the E.K.IV.

Load from disk: Load a saved configuration program from local disk into the configuration panel.

Save to disk: Save the actual configuration on a local disk as an extension file «.nnz».

F. Configuration / control panel selection

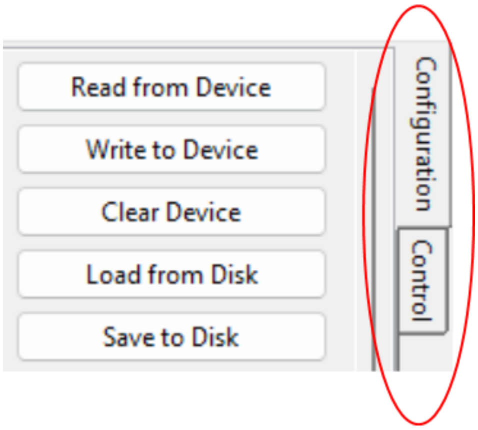

Figure 10: Configuration/control panel selection.

This button switches between the configuration and the control windows.

Warning: To ensure proper operation and to prevent damage to the sensors, the maximum voltages of the sensors and heating elements to be applied must not exceed:

	Heater voltages (VH)	Sensor voltages (Vs)
NZGS2	2.2 V	+/-0.8V
NZGS3	1.4V	+/-0.5V

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G. Summary of the operation procedure

Now that whole operating procedures has been explained, here is the summary:

- Launch «Nanoz Config»
- Plug Nanoz and environmental sensors to the evaluation kit
- Plug the evaluation kit to the computer
- Identify the right COM then press connect
- Press read ***from device***
- Define the number of cycles and sequences
- For each cycle, define the order of execution of sequences
- Define the parameters of each sequence
- Press ***Save to disk*** (optional)
- Press **Write to device**

Note: At this point, the E.K.IV is ready to execute the defined cycles and sequences.

V. Control window

The control window is used to execute cycles and sequences written on the E.K.IV and to visualize the sensor, heater, gas concentration and environmental signals. It can be divided in 5 setting panels as described in Table 3 and Figure 11.

	Name
A	Cycle and graphic settings
B	Sensor signals display
C	Heater signals display
D	Environmental signals display
E	Configuration / Control panel selection

Table 3 : Setting panels of the "Control" window.

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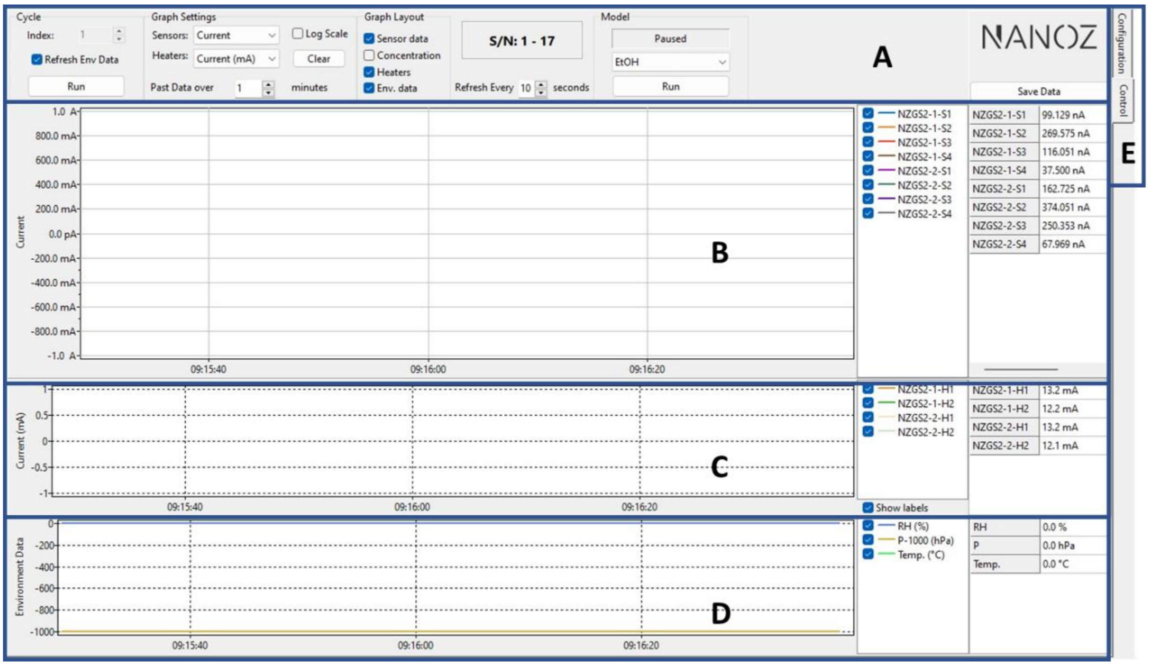

Figure 11: Setting panels in the "Control" window.

A. Control panel settings

Figure 12 : Cycle and graphic setting.

Cycle:

- Index: Used to select the cycle to execute.
- Refresh Env Data**:** If checked, Data in the «Environment data» (Temperature, Relative Humidity and atmospheric pressure) are always displayed, without cycle running.

Graph settings:

- Sensors: Choose the sensor characteristic to display (Current or Resistance).

Note: The display can be done on a logarithmic scale by clicking on the "log scale" checkbox.

- Heaters: Choose the heater characteristic to display (Voltage, Resistance, Current or Power).
- Clear: Delete all displayed signals.
- Past Data over: Define display time of the curves in minutes (x-axis).
- Log Scale: When checked, the logarithmic scale will be applied on the sensor current graph.

Graph Layout:

- Allows you to choose the graph to display.

Model:

- This option can be used in case a gas prediction model is provided to the user. Otherwise, please ignore it.

Run: Run the cycle selected in the **index** (here cycle 1).

<u>Note</u>*:* Initial current levels are highly dependent on the surrounding environment.

Save Data: Press «save data» to start recording data.

<u>Note</u>: Data will be saved in the same folder containing the EK software.

Data will be saved in «.CSV» format and can be opened with Microsoft Excel or equivalent programs.

<u>Note:</u> In total, two files will be saved named: ENV and SPL.

- *The first file «ENV» contains the Environment data (Temperature (°C), Relative Humidity (RH%) and Pressure (P-1000 (hPa))).*
- *The second file «SPL» contains the following information: date of the test, time of the test, chip number, current (µA), voltage (mV).*

<u>Note:</u> When a cycle is running, parameters in the "Configuration window" cannot be changed.

<u>Note:</u> Sensor, heater and environmental characteristics are displayed in real time in the tables at the right side of each graph.